

Research Statement

University of Tokyo Research Internship Programme (UTRIP) -- Kavish Kumar

Applicant: Kavish Kumar
Institution: Plaksha University, Mohali
Programme: BTech -- Computer Science and Artificial Intelligence
Email: kavish.kumar.ug23@plaksha.edu.in

My research interest lies at the intersection of machine learning and human-centred computing -- specifically, building models that are robust to real-world variation in human behaviour and biology. I am applying to UTRIP to deepen this work in a rigorous, internationally collaborative environment.

My most significant project to date is an automatic speech recognition system for dysarthric speakers, developed at Plaksha University. Dysarthria -- caused by conditions such as ALS and Cerebral Palsy -- distorts speech in ways that render standard ASR systems unreliable. I built a speaker-independent system using the wav2vec 2.0 model to extract latent acoustic embeddings, feeding these into a bidirectional LSTM for sequence prediction. I pre-trained on LibriSpeech and fine-tuned on the TORGO dataset, achieving a word error rate of 48% and a character error rate of 27% -- approaching the accuracy of speaker-dependent systems, which require individual calibration per user. This result demonstrated that generalised dysarthric ASR is tractable, and surfaced several questions I want to pursue: how do self-supervised representations handle non-standard phonation, and can we transfer across dysarthria subtypes with limited labelled data?

Alongside this, I have worked on predictive modelling with structured data. My flight delay analysis project involved large-scale web scraping of live FlightRadar24 data, XGBoost classification of delay events, and a logistic regression model that isolated airline-specific delay contribution -- finding that switching from Air India to IndiGo reduces delay odds by 84% domestically. At NTU Singapore in the summer of 2024, I applied k-means clustering and random forest regression to box office prediction, gaining experience in collaborative, cross-institutional research.

At UTRIP, I am particularly interested in joining a group working on computational biology, machine learning for scientific data, or human-computer interaction. I want to understand how methods from deep learning -- especially representation learning -- can be adapted to domains where labelled data is scarce and individual variation is high. I am comfortable in Python, experienced with PyTorch and the HuggingFace ecosystem, and eager to contribute to ongoing lab work from day one.

I see UTRIP as an opportunity not just to advance a specific project, but to develop the research instincts -- how to frame a problem, design a controlled experiment, and communicate findings -- that will define the rest of my academic career.